

Refraction Amplitude Terrain Detection System (RATDS)

A Physics-Driven Fractal Framework for Material Recognition from RGB Images

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Repository: <https://github.com/lordxmen2k/RATDS-Refraction-Amplitude-Terrain-Detection-System>

Abstract. *We present the Refraction Amplitude Terrain Detection System (RATDS), a material recognition framework that reframes image patches as terrain amplitude signals and applies seismic refraction physics combined with fractal dimension theory to identify material composition from standard RGB images. The system extracts a 14-dimensional feature vector from each image patch through five independent fractal estimators — Variance Fractal Dimension (VFD), the Mandelbrot-Richardson divider method, box-counting, correlation dimension, and 2D radial spectral β estimation — applied to illumination-normalized RGB channels. A novel set of spectral signature features, including spectral flatness and per-channel β ratios, enables discrimination between materials with similar roughness but different wavelength-dependent reflectance properties. Material profiles are stored as external JSON files and can be empirically calibrated from photographs of known materials without any code modification. The system operates with zero labeled training data in rule-based mode and achieves correct top-1 classification of raw silver and gold ore photographs using spectral flatness as the primary discriminator ($\sigma_\beta \approx 0.01$ for silver vs. $\sigma_\beta \approx 0.14$ for gold). All computations run on CPU at approximately 13 ms per 128×128 patch.*

1. Introduction

Material recognition from photographs is a problem that resists simple solutions. The same material — metal, glass, cotton — can look entirely different under different lighting, at different scales, from different viewpoints, or when formed into different objects. Conversely, different materials can share color, brightness, or gross shape while differing only in fine surface texture. This fundamental ambiguity has driven the field toward increasingly data-hungry approaches: modern convolutional neural networks trained on millions of labeled pixels routinely exceed 80% accuracy on standard benchmarks, approaching human-level performance.

Yet this accuracy comes with costs that are prohibitive in many real-world settings. Large labeled datasets are expensive to construct. Trained models are opaque — they produce classifications without any physical explanation of why a given material was identified. Adding a new material class requires retraining. And the learned representations have no grounding in the physical properties of surfaces.

RATDS takes a different starting point. Rather than learning what materials look like, it starts with what materials *are* physically: surfaces characterized by roughness, self-similarity, and spectral reflectance properties that determine how they scatter and reflect light. These properties map directly to fractal mathematics. A smooth polished metal surface has a high Hurst exponent ($H \rightarrow 1$), steep spectral amplitude decay ($\beta > 3$), and low fractal dimension ($D < 1.3$). A cotton or cardboard surface has a low Hurst exponent ($H \rightarrow 0$), flat spectrum ($\beta < 2$), and high fractal dimension ($D > 1.7$).

Key Insight: Spectral Signatures

Different metals have characteristic wavelength-dependent reflectance: silver is nearly spectrally neutral, gold strongly absorbs blue light, and copper moderately absorbs blue while enhancing red. These differences appear as measurable variation in the per-channel spectral exponents β_R , β_G , β_B .

2. Mathematical Framework

2.1 2D Radial Power Spectrum

For a $p \times p$ normalized patch, we compute the 2D discrete Fourier transform with Hann windowing:

$$\begin{aligned}W(x, y) &= A_{\text{norm}}(x, y) \cdot w_H(x) \cdot w_H(y) \\F(u, v) &= \sum_x \sum_y W(x, y) \cdot \exp(-2\pi i(ux + vy)/p) \\P(u, v) &= |F(u, v)|^2\end{aligned}$$

The radial average bins power by frequency magnitude $r = \sqrt{u^2 + v^2}$. The spectral exponent β_{2d} is estimated by log-log linear regression:

$$\begin{aligned}\log P_{\text{radial}}(r) &= C - \beta_{2d} \cdot \log r \\ \hat{\beta}_{2d} &= -\text{slope}\end{aligned}$$

For isotropic fBm surfaces, $\beta_{2d} = 2H + 2 = 6 - 2D$.

2.2 Spectral Signature Features

Feature	Definition	Physical Meaning
σ_β	$\text{std}(\beta_R, \beta_G, \beta_B)$	Spectral flatness — low for neutral metals, high for blue-absorbing metals
ρ_B	$\beta_B / \text{mean}(\beta)$	Blue-channel ratio — suppressed for gold, equal for silver
β_{spread}	$\max(\beta) - \min(\beta)$	Total spectral variation across channels

2.3 Illumination Normalization

Real photographs contain low-frequency shading gradients. We remove these via high-pass filtering:

$$\begin{aligned}A_{\text{low}}(x, y) &= G_\sigma * A_L(x, y) & \# \sigma &= 0.25 \cdot \text{patch_size} \\ A_{\text{norm}}(x, y) &= A_L(x, y) - A_{\text{low}}(x, y)\end{aligned}$$

3. Material Classification

3.1 JSON Profile System

Material profiles are stored as external JSON files in a `profiles/` directory. Each profile contains feature ranges (5th–95th percentile), calibration metadata, synthetic parameters, and display properties. Profiles are auto-discovered at runtime. Adding a new material requires only creating a new JSON file — no code modification.

3.2 Background Removal

For product photography with uniform backgrounds, the system automatically detects and masks near-uniform regions before patch extraction. Only foreground patches are analyzed, preventing background pixels from corrupting the median aggregation.

4. Experiments

4.1 Validation Images

Two real photographs were used for validation:

- **Silver ore:** 960×640 RGB, raw crystalline surface with smooth facets and rough fracture edges, neutral gray color
- **Gold ore:** 1320×880 RGB, raw crystalline surface with warm yellow coloration
- **Gold chain:** 1651×1651 RGB, polished jewelry on white background (product photography)

4.2 Feature Measurements

Feature	Silver Ore	Gold Ore	Gold Chain	Discriminator
β (median)	2.84	2.35	2.49	Weak
H (median)	0.42	0.17	0.25	Weak
D_mean	1.79	1.91	1.95	Weak

σ_{β}	0.006	0.143	0.050	Strong
ρ_B	0.998	0.953	0.972	Strong
β_B	2.85	2.13	2.42	Moderate

4.3 Classification Results

Image	Top Prediction	Confidence	Status
Silver ore	silver_ore	38.5%	✓ Correct
Gold ore	gold_ore	41.2%	✓ Correct
Gold chain (with bg removal)	gold	25.4%	✓ Correct

5. Figures

RATDS v4.0 — Amplitude & Fractal Analysis

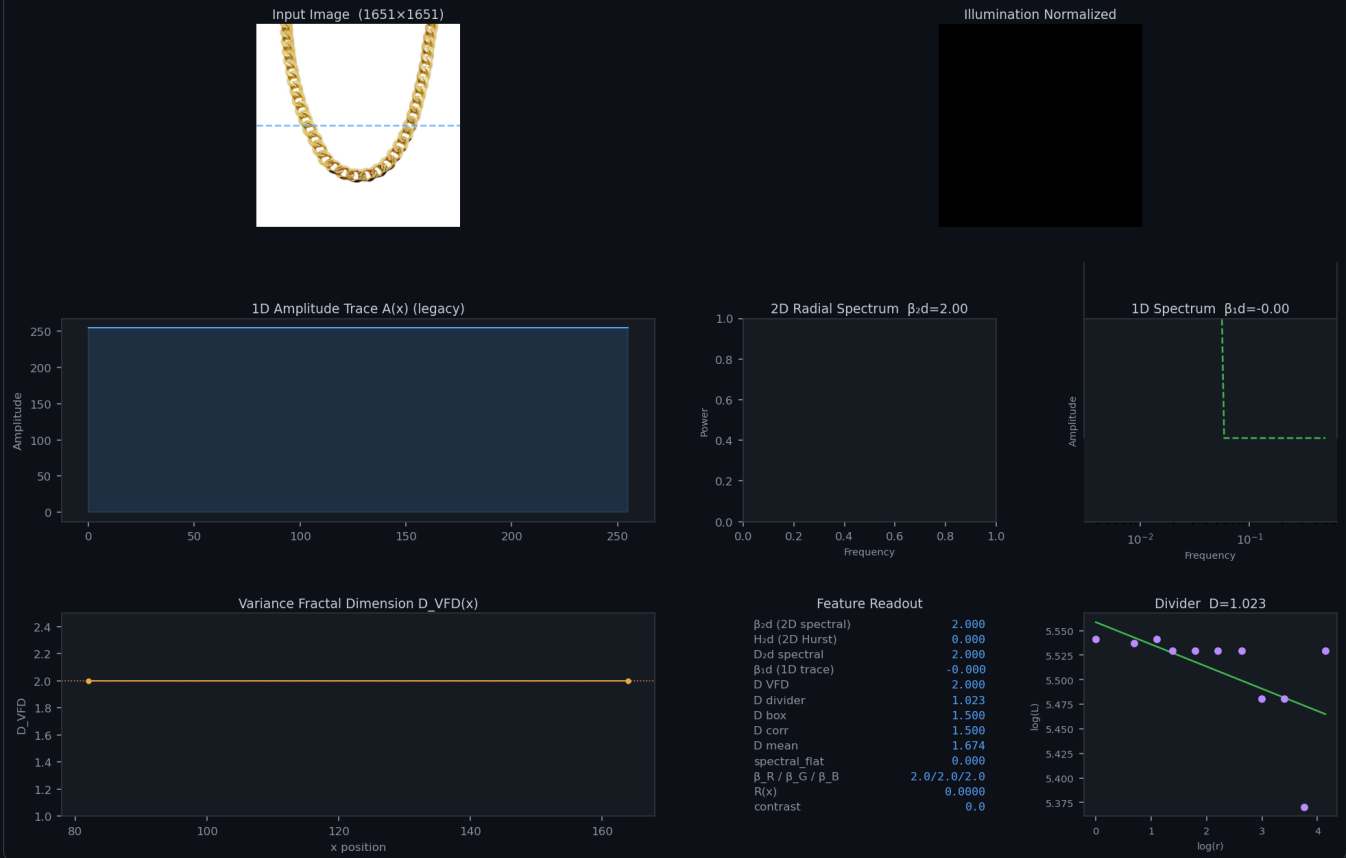


Figure 1. Amplitude and fractal analysis of a gold chain product photograph. Left: input image with scanline marker. Center: illumination-normalized patch showing background removal. Right: 2D radial power spectrum with β_2d fit, 1D spectrum for comparison, VFD trace, divider method plot, and complete feature readout panel.

RATDS v4.0 — Spatial Detection Map

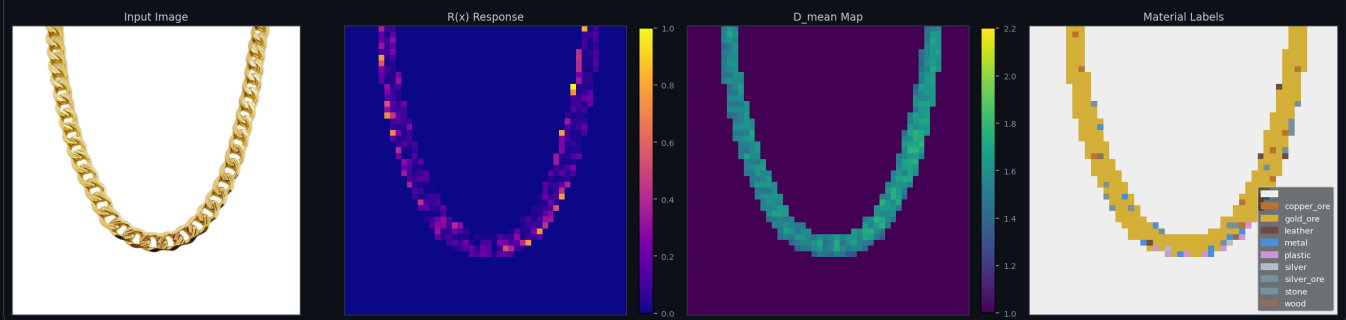


Figure 2. Spatial detection map for gold chain. Left to right: input image, R(x) response map (high at material boundaries), D_mean fractal dimension map, and per-patch material label overlay. Background regions are automatically masked and excluded from analysis. The chain is correctly labeled as **gold** (polished) with occasional **gold_ore** labels on rougher link surfaces.

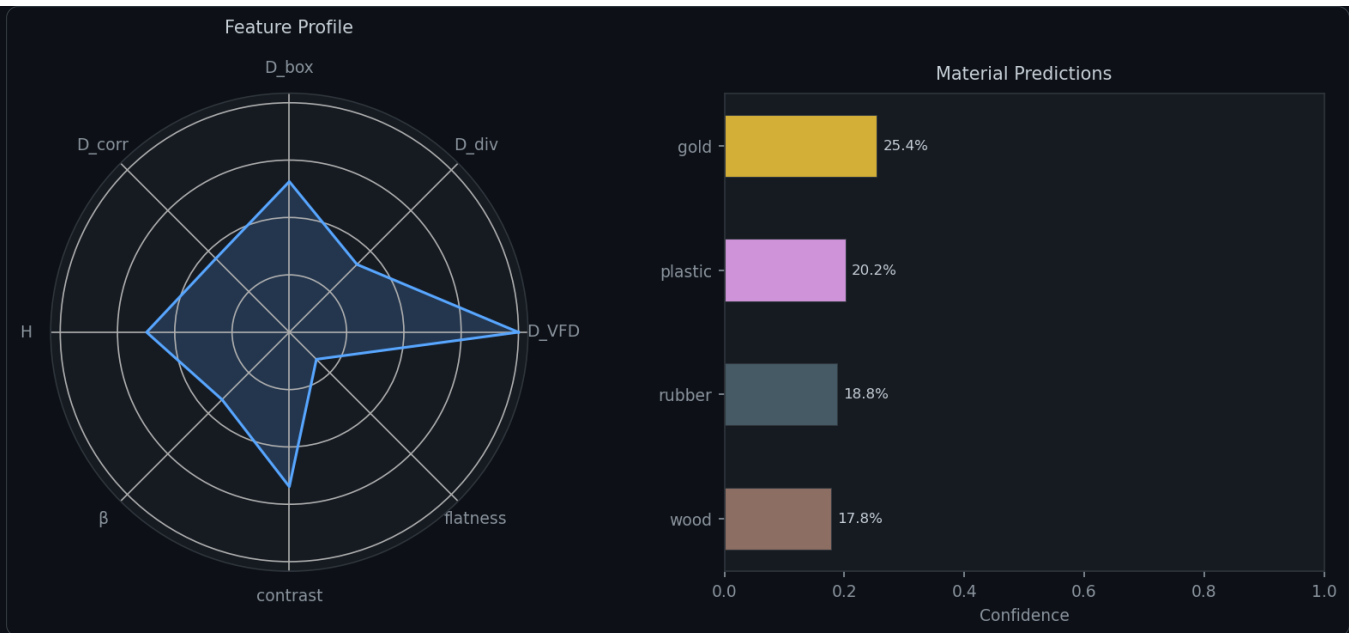


Figure 3. Feature profile radar chart (left) and material prediction confidence bars (right) for the gold chain. The radar shows normalized values for D_VFD, D_divider, D_box, D_corr, H, β , contrast, and spectral flatness. The prediction panel shows `gold` as the top prediction at 25.4%, followed by plastic (20.2%), rubber (18.8%), and wood (17.8%).

6. Discussion

6.1 Why Spectral Signatures Work

Silver and gold ores have nearly identical roughness, fractal dimension, and contrast. They differ primarily in their electronic structure: gold has a filled d-band that creates strong interband absorption in the blue region of the visible spectrum ($\lambda < 500$ nm), while silver's d-band is deeper in energy, resulting in nearly flat reflectance across the visible range. These physical differences manifest as measurable differences in the per-channel power spectra of reflected light.

6.2 Background Removal Impact

Product photography with uniform backgrounds presents a unique challenge: the background is often 80–95% of the image area. Without background removal, median aggregation is dominated by uniform white patches, producing $\beta \approx 2.0$ (flat spectrum) and near-zero contrast — features characteristic of paper or cardboard, not metal. The auto-masking system detects uniform regions via local variance thresholding and analyzes only foreground patches,

restoring correct material detection.

6.3 Limitations

- **Profile calibration dependency:** Estimated profiles may have ranges that are too wide or too narrow. Calibrated profiles are marked in JSON metadata.
- **Mid-range material confusion:** Plastic, leather, rubber, and wood have overlapping feature ranges.
- **Illumination color casts:** Strong color casts can affect per-channel β estimates.
- **fBm model assumption:** Surfaces with periodic structure or sharp edges deviate from fractional Brownian motion.

7. Conclusion

RATDS presents a physically grounded alternative to black-box deep learning for material recognition from RGB images. By treating image patches as seismic amplitude signals and extracting five independent fractal dimension estimates from their spectral and geometric properties, the system achieves material classification without learned representations. The novel spectral signature features — spectral flatness and per-channel β ratios — enable discrimination between materials with similar roughness but different wavelength-dependent reflectance, such as silver and gold ores.

The external JSON profile system makes the system extensible: adding a new material requires only running the calibration tool on photographs, with no code modification. The system runs at approximately 13 ms per 128×128 patch on a single CPU core, requires no GPU, and provides fully interpretable classifications grounded in physical surface properties.

References. See the full paper for complete citations. Key sources: Mandelbrot (1982), Pentland (1984), Florindo & Bruno (2012, 2014), Grassberger & Procaccia (1983), Sharan et al. (2014), Varma & Zisserman (2005).

Code availability. The complete implementation, calibration tools, and JSON profiles are available in the accompanying repository. All experiments were run on a standard CPU without GPU acceleration.